RoveFlight S3 / 1S brushed quadcopter.
A.0 engineering prototype: see docs/Manufacturing_Readiness.md before fabrication.
Protected 1S battery; external charger. USB forces motor disarm.
Global nets join functional sheets. Four-layer board; 45 x 45 mm nominal.

GND
BAT_RAW
BAT_FUSED
BAT_PROTECTED
VM
SYS_IN
+3V3_SENS
REG_VINA
IR_VS

01 ESP32S3 Controller 1
File: 01_ESP32S3_Controller_1.kicad_sch

02 USB Programming 1
File: 02_USB_Programming_1.kicad_sch

03 Battery Power 1
File: 03_Battery_Power_1.kicad_sch

04 Motor Safety Interlock 1
File: 04_Motor_Safety_Interlock_1.kicad_sch

05 Motor Safety Interlock 2
File: 05_Motor_Safety_Interlock_2.kicad_sch

06 Motor M1 1
File: 06_Motor_M1_1.kicad_sch

07 Motor M2 1
File: 07_Motor_M2_1.kicad_sch

08 Motor M3 1
File: 08_Motor_M3_1.kicad_sch

09 Motor M4 1
File: 09_Motor_M4_1.kicad_sch

10 SPI Flight Sensors 1
File: 10_SPI_Flight_Sensors_1.kicad_sch

11 Optical and IR 1
File: 11_Optical_and_IR_1.kicad_sch

12 Battery Monitoring 1
File: 12_Battery_Monitoring_1.kicad_sch

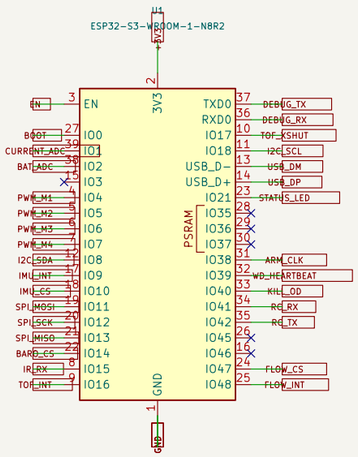
13 Receiver Expansion and Indicators 1
File: 13_Receiver_Expansion_and_Indicators_1.kicad_sch

14 Receiver Expansion and Indicators 2
File: 14_Receiver_Expansion_and_Indicators_2.kicad_sch

Sheet: /
File: RoveFlight_S3.kicad_sch

Title: RoveFlight S3 | RF-S3-A0

Size: A3	Date: 2026-11-08	**Rev: A.0 ENGINEERING PROTOT**
KiCad E.D.A. 10.0.6		Id: 1/15







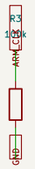












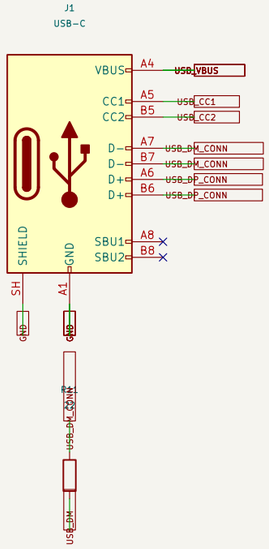


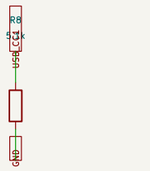
N8R2: 8MB Quad SPI flash, 2MB Quad PSRAM. Do not substitute Octal PSRAM variants.
GPIO19/20 dedicated USB. GPIO0/3/45/46 are straps; GPIO35/36/37 unused.
GPIO38 arming clock; GPIO39 watchdog; GPIO40 open-drain KILL. Motor PWM GPIO4..7.

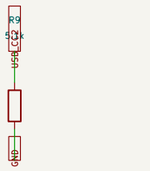
Sheet: /01 ESP32S3 Controller 1/
File: 01_ESP32S3_Controller_1.kicad_sch

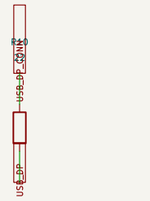
Title: RoveFlight S3 | ESP32S3 Controller

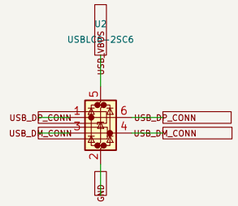
Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 2/15

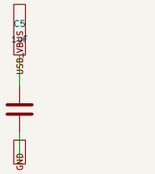










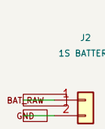


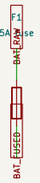
Native ESP32-S3 USB Serial/JTAG, no UART bridge. USB VBUS powers logic through diode OR.
USB insertion independently asserts the motor latch clear. No battery charging.

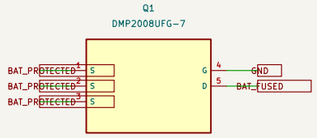
Sheet: /02 USB Programming 1/
File: 02_USB_Programming_1.kicad_sch

Title: RoveFlight S3 | USB Programming

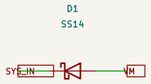
Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 3/15







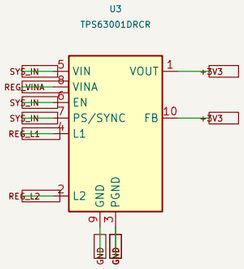


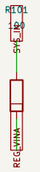






























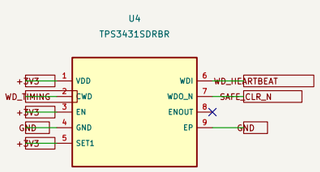


1S LiPo 3.1–4.2V; 4A continuous aggregate motor current, stall pulses separately qualified.
Q1 reverse-polarity protection, F1 catastrophic fault backup; EXTERNAL protected pack required.
TPS63001 fixed 3V3, forced PWM. SYS_IN diode OR cannot charge battery. No charger fitted.

Sheet: /03 Battery Power 1/
File: 03_Battery_Power_1.kicad_sch

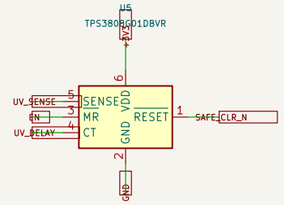
Title: RoveFlight S3 | Battery Power

Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 4/15

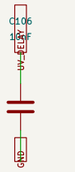




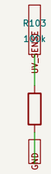




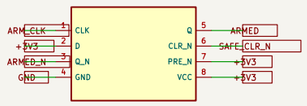




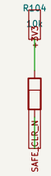




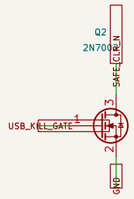






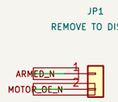














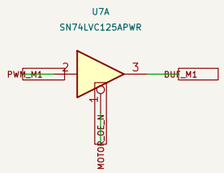


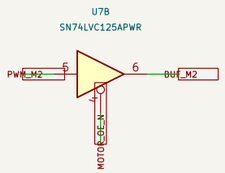
SAFE_CLR_N is wired-OR: watchdog, VM undervoltage, USB, KILL button, firmware OD.
Async clear latches disarmed. A NEW rising ARM_CLK is required after every fault.
JP1 physically disconnects enable. PWM gates high impedance when disarmed, with 10k gate pulldowns.
Watchdog 1nF: 132.4ms nominal, falling-edge heartbeat <20ms required from healthy control task.

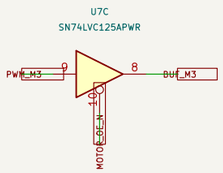
Sheet: /04 Motor Safety Interlock 1/
File: 04_Motor_Safety_Interlock_1.kicad_sch

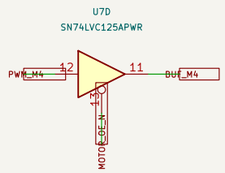
Title: RoveFlight S3 | Motor Safety Interlock

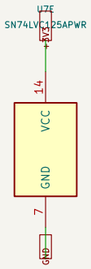
Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 5/15









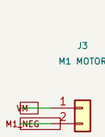


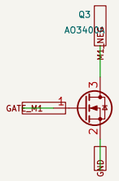
SAFE_CLR_N is wired-OR: watchdog, VM undervoltage, USB, KILL button, firmware OD.
Async clear latches disarmed. A NEW rising ARM_CLK is required after every fault.
JP1 physically disconnects enable. PWM gates high impedance when disarmed, with 10k gate pulldowns.
Watchdog 1nF: 132.4ms nominal, falling-edge heartbeat <20ms required from healthy control task.

Sheet: /05 Motor Safety Interlock 2/
File: 05_Motor_Safety_Interlock_2.kicad_sch

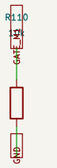
Title: RoveFlight S3 | Motor Safety Interlock

Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 6/15













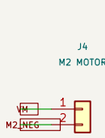


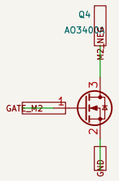
15 BRUSHED ONLY. Pin1 VM+, pin2 switched motor negative. 1A running / 2A brief stall target.
100R gate resistor; 10k gate-source pulldown; SS24 flyback cathode VM, anode drain.
Fit additional 100nF directly at motor terminals; twist leads. Rotation is set by motor wiring.

Sheet: /06 Motor M1 1/
File: 06_Motor_M1_1.kicad_sch

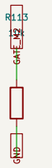
Title: RoveFlight S3 | Motor M1

Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 7/15













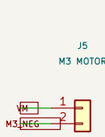


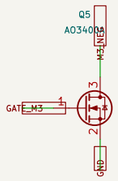
15 BRUSHED ONLY. Pin1 VM+, pin2 switched motor negative. 1A running / 2A brief stall target.
100R gate resistor; 10k gate–source pulldown; SS24 flyback cathode VM, anode drain.
Fit additional 100nF directly at motor terminals; twist leads. Rotation is set by motor wiring.

Sheet: /07 Motor M2 1/
File: 07_Motor_M2_1.kicad_sch

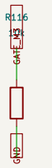
Title: RoveFlight S3 | Motor M2

Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 8/15













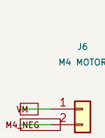


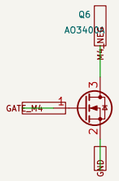
15 BRUSHED ONLY. Pin1 VM+, pin2 switched motor negative. 1A running / 2A brief stall target.
100R gate resistor; 10k gate–source pulldown; SS24 flyback cathode VM, anode drain.
Fit additional 100nF directly at motor terminals; twist leads. Rotation is set by motor wiring.

Sheet: /08 Motor M3 1/
File: 08_Motor_M3_1.kicad_sch

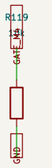
Title: RoveFlight S3 | Motor M3

Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 9/15













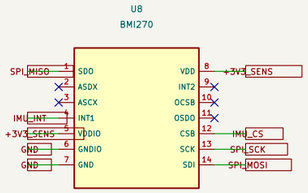


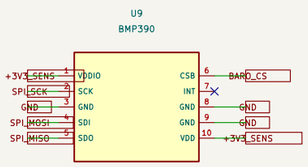
15 BRUSHED ONLY. Pin1 VM+, pin2 switched motor negative. 1A running / 2A brief stall target.
100R gate resistor; 10k gate–source pulldown; SS24 flyback cathode VM, anode drain.
Fit additional 100nF directly at motor terminals; twist leads. Rotation is set by motor wiring.

Sheet: /09 Motor M4 1/
File: 09_Motor_M4_1.kicad_sch

Title: RoveFlight S3 | Motor M4

Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 10/15













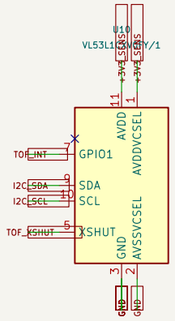


BMI270 primary 4-wire SPI, INT1 data ready. Auxiliary/OIS pins intentionally NC.
BMP390 on shared SPI with independent CS; INT unused. Pressure port faces underside.
No compass: motor conductors are too close for a defensible uncalibrated onboard heading.

Sheet: /10 SPI Flight Sensors 1/
File: 10_SPI_Flight_Sensors_1.kicad_sch

Title: RoveFlight S3 | SPI Flight Sensors

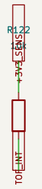
Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 11/15





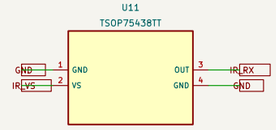












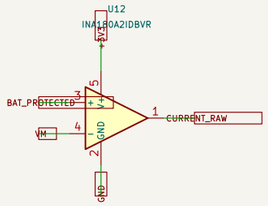




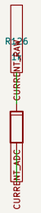


VL53L1X faces DOWN, I2C 0x29. Firmware must enable 2V8 I/O mode for 3V3 supply.
Use near-ground ranging only; sunlight, reflective floors, tilt and prop wash limit reliability.
TSOP75438TT is a 38kHz receiver, not proximity sensing. Optical flow is an external SPI module.

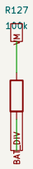
Sheet: /11 Optical and IR 1/		
File: 11_Optical_and_IR_1.kicad_sch		
Title: RoveFlight S3 | Optical and IR		
Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
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Kelvin sense R1, 5mOhm. INA180A2 gain50: 0.25V/A. ADC1 GPIO1.
Battery ADC1 GPIO2, 1:2 divider, RC filter. Calibrate ADC offset/gain in firmware.
Fuse is backup short-circuit protection; firmware must implement current and low-battery shutdown.

Sheet: /12 Battery Monitoring 1/
File: 12_Battery_Monitoring_1.kicad_sch

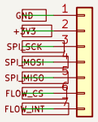
Title: RoveFlight S3 | Battery Monitoring

Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
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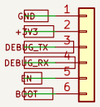
J7
RC 3V3



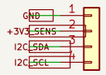
J8
OPTICAL FLOW SPI



J9
DEBUG UART



J10
I2C 3V3



D7
GREEN



R130
470R
STATUS_LED
LED_A

TP1
+3V3

TP2
GND

TP3
VM

TP4
I2C_SDA

TP5
I2C_SCL

TP6
SPI_SCK

TP7
SPI_MOSI

TP8
SPI_MISO

TP9
DEBUG_TX

TP10
DEBUG_RX

TP11
PWM_M1

TP12
PWM_M2

TP13
PWM_M3

TP14
PWM_M4

Receiver: 3V3 UART CRSF/IBUS only, no 5V tolerance or onboard SBUS inverter.
External PMW3901 breakout must include its required regulator/level compatibility and lens.
SPI test pads/flow connector are 3V3; sensor SPI max10MHz, initial firmware 1MHz.

Sheet: /13 Receiver Expansion and Indicators 1/
File: 13_Receiver_Expansion_and_Indicators_1.kicad_sch

Title: RoveFlight S3 | Receiver Expansion and Indicators

Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
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TP15
SAFE_CLR_N

TP16
ARMED

Receiver: 3V3 UART CRSF/IBUS only, no 5V tolerance or onboard SBUS inverter.
External PMW3901 breakout must include its required regulator/level compatibility and lens.
SPI test pads/flow connector are 3V3; sensor SPI max10MHz, initial firmware 1MHz.

Sheet: /14 Receiver Expansion and Indicators 2/
File: 14_Receiver_Expansion_and_Indicators_2.kicad_sch

Title: RoveFlight S3 | Receiver Expansion and Indicators

Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 15/15